

3.5 Random Variables

The variable that connects the outcome of a Random Experiment to a real number.

Example: 'x' is the value of the number that a dice shows when it is rolled.

Discrete RV → The RV whose value is obtained by counting, defined by PMF

Random Variable

Continuous RV → The RV whose value is obtained by Measuring, defined by PDF

- If a data consists of ' f_1 ' data points with value ' x_1 ', ' f_2 ' data points with value ' x_2 ' ' f_n ' data point with value ' x_n ', then
 - Expectation of 'x' = $E(x) = \sum_{i=1}^n x_i P(x = x_i)$
 - Variance of 'x' = $\sigma^2 = E(x^2) - (E(x))^2$ and σ is the standard deviation.

3.5.1 Probability Mass Function (PMF)

The PMF $p(x)$ of a discrete random variable X taking values $x_1, x_2, \dots, x_n$ is defined such that,

- $p(x_i) \geq 0$
- $\sum_{i=1}^n p(x_i) = 1$
- $p(x_i) = p(X = x_i)$

3.5.2 Probability Density Function (PDF)

The pdf $f(x)$ of a continuous random variable X is defined such that,

- $f(x) \geq 0$
- $\int_{-\infty}^{\infty} f(x) dx = 1$
- $P(a < X < b) = \int_a^b f(x) dx$

3.5.3 Expected Value

- Expected value of a random variable X, $E[X]$, is defined as, $E[X] = \begin{cases} \sum xp(x); & X \text{ is discrete rv} \\ \int_{-\infty}^{\infty} xf(x)dx; & X \text{ is continuous rv} \end{cases}$
- Expected value of X^2 is,

$$E[X^2] = \begin{cases} \sum x^2 p(x); & X \text{ is discrete rv} \\ \int_{-\infty}^{\infty} x^2 f(x) dx; & X \text{ is continuous rv} \end{cases}$$

Note: $E[X^n]$ is called nth moment.

3.5.4 Mean of Random Variable 'X'

$$\text{Mean} = \mu = E[X]$$

3.5.5 Variance of a Random Variable 'X'

$$\text{Var}(X) = E[(X - \mu)^2]$$

$$\text{Or, Var}(X) = E[X^2] - \mu^2$$

3.5.6 Properties of Expectation

- (i) $E[c] = c$, c is a constant.
- (ii) $E[ax] = aE[X]$
- (iii) $E(aX + b) = aE(X) + b$
- (iv) If X and Y are random variable $E[X \pm Y] = E(x) \pm E(Y)$.
- (v) If X and Y are random variables $E(X \cdot Y) = E(X) \cdot E(Y / X)$.
- (vi) If X and Y independent random variables $E(X \cdot Y) = E(X) \cdot E(Y)$.

3.5.7 Properties of Variance

- (i) $\text{Var}[C] = 0$, C is constant.
- (ii) $\text{Var}(aX) = a^2 \text{Var}(X)$ where X is random variable and ' a ' constant.
 $\text{Var}(-X) = (-1)^2 \text{Var}(X) = \text{Var}(X)$ Variance is always positive.
- (iii) $\text{Var}(ax + b) = a^2 \text{Var}(X) + 0$
- (iv) If X and Y are independent random variables.
 $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$
 $\text{Var}(X - Y) = \text{Var}(X) + \text{Var}(Y)$
- (v) $\text{Var}(ax + by) = a^2 \text{Var}(x) + b^2 \text{Var}(y) + 2ab \text{Cov}(x, y)$
- (vi) $\text{Cov}(x, y) = E(x, y) - E(x) E(y)$
- (vii) For independent random variables $\text{Cov}(x, y) = 0$

3.5.8 Continuous RV

The value of the Random Variable is obtained by Measuring.

3.6 Probability Distribution Function (PDF)

A continuous & differentiable function $P(x)$ is said to be a probability distribution/density function of a continuous random variable ' x ' if $P(a \leq x \leq b) = \int_a^b P(x)dx$

3.6.1 Mean (or) Expectation

If $P(x)$ is a probability distribution/density function of a continuous Random Variable ' x ' then the Mean of ' x ' = $E(x) = \int_{-\infty}^{\infty} x \cdot P(x) dx$

3.6.2 Median

The value of ' x ' for which the total probability is exactly divided into two equal halves is called Median.

3.6.3 Mode

The value of ' x ' at which $P(x)$ is maximum is called mode.

3.6.4 Variance

$$= \sigma^2 = E(x^2) - (E(x))^2$$

$$\Rightarrow \sigma^2 = \int_{-\infty}^{\infty} x^2 \cdot P(x) dx - \left(\int_{-\infty}^{\infty} x \cdot P(x) dx \right)^2$$

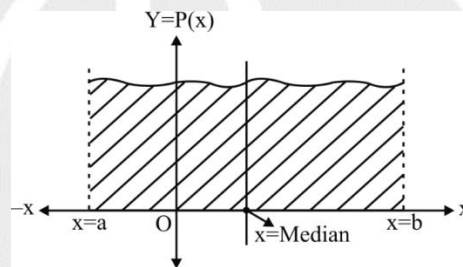


Fig. 5.7. Continuous random variables

3.7 Continuous RV distributions

(1) Gaussian/Normal Distribution:

If ' x ' is a continuous Random variable with mean ' μ ' and standard deviation ' σ ', then the probability distribution/density function of normally distributed variable ' x ' is given by

$$P(x) = \frac{1}{\sigma\sqrt{2\pi}} \cdot e^{\left(\frac{-(x-\mu)^2}{2\sigma^2}\right)}$$

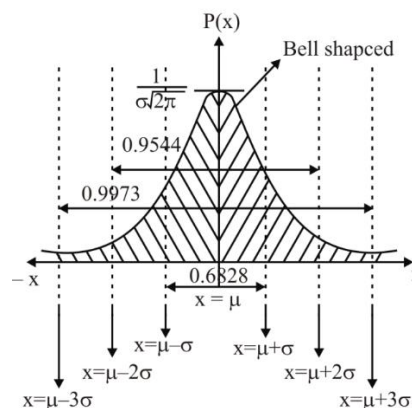


Fig. 5.8. Normal distribution

$$\text{Mean} = \text{Median} = \text{Mode} = \mu$$

$$P(\mu - \sigma \leq x \leq \mu + \sigma) = 0.6828$$

$$P(\mu - 2\sigma \leq x \leq \mu + 2\sigma) = 0.9544$$

$$P(\mu - 3\sigma \leq x \leq \mu + 3\sigma) = 0.9973$$

$$P(x) = \frac{1}{\sigma \cdot \sqrt{2\pi}} \cdot e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

(2) Standard Normal Distribution:

$$\text{Assuming } z = \frac{x-\mu}{\sigma}; \mu = 0; \sigma = 1, P(z) = \frac{1}{\sqrt{2\pi}} \cdot e^{-\frac{z^2}{2}}$$

$$P(-1 \leq z \leq 1) = 0.6828$$

$$P(-2 \leq z \leq 2) = 0.9544$$

$$P(-3 \leq z \leq 3) = 0.9973$$

Note:

1. The normal distribution curve is bell shaped curve
2. The points of inflection of the normal distribution curve are at $x = \mu + \sigma$ and $x = \mu - \sigma$.
3. The cumulative function graph is of 'S' Shape.
4. For a given normal distribution, Mean = median = Mode

(3) Uniform Distribution:

If 'x' is a uniformly distributed random variable such that $a \leq x \leq b$ then the Pdf is given by

$$P(x) = \frac{1}{(b-a)}$$

$$\text{Mean} = \int_a^b x \cdot P(x) dx = \int_a^b x \cdot \frac{1}{b-a} dx = \frac{1}{(b-a)} \int_a^b x \cdot dx$$

$$\left(\frac{b+a}{2} \right) = \text{Mean}$$

$$\Rightarrow \text{Variance} = \sigma^2 = \frac{(b-a)^2}{12}$$

$$\text{Std. deviation} = \sigma = \frac{(b-a)}{\sqrt{12}}$$

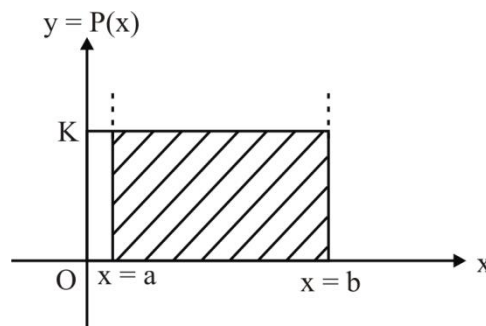


Fig. 5.9. Uniform Distribution

3.7.1 Properties of Mean and Variance:

$$E(ax + by) = a \cdot E(x) + b \cdot E(y)$$

$$V(ax + by) = a^2 \cdot V(x) + b^2 \cdot V(y) - 2abCOV(x, y)$$

$$\text{where } COV(x, y) = E(xy) - E(x) \cdot E(y)$$

If x, y are independent random variables, then $E(xy) = E(x) \cdot E(y) \Rightarrow COV(x, y) = 0$

(1) Exponential Distribution:

If ' x ' is a continuous random variable with mean as $\frac{1}{\lambda}$ then the exponential distribution of ' x ' is

given by the function

$$f(x) = \begin{cases} \lambda \cdot e^{-\lambda x} & ; x \geq 0 \\ 0 & : otherwise \end{cases}$$

$$\text{Mean} = \frac{1}{\lambda}$$

$$\sigma^2 = \frac{1}{\lambda^2}$$

Mean = Standard Deviation = $\frac{1}{\lambda}$

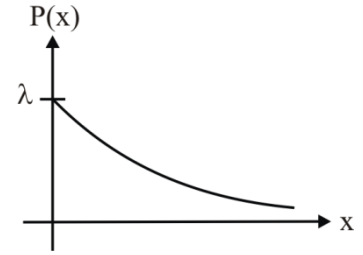


Fig. 5.10. Exponential distribution

3.8 Discrete Random Variable Distributions

If a Random experiment has **only two Possible outcomes**, (one is Success & other is failure) and the Probability of Success doesn't depend on time, then the probability of occurring of exactly ' r -successes' in ' n -trials' is given by

$$P(X = r) = {}^n C_r \cdot p^r \cdot q^{n-r}$$

Where, $p \rightarrow$ Probability of Success,

$q \rightarrow$ Probability of Failure

$$p + q = 1$$

Mean = np , Variance = $npq = \sigma^2$, standard deviation = $\sigma = \sqrt{npq}$

3.8.1 Poisson Distribution

If a random experiment has only two possible outcomes, and the average number of successes in a given time ' t ' is λ , then the probability that exactly ' r ' successes occur within the same time ' t ' given by

$$P(x = r) = \frac{e^{-\lambda} \cdot \lambda^r}{r!}$$

$$\text{Mean} = \lambda.$$

$$\text{Mean} = \text{Variance} = \lambda$$

$$\Rightarrow \sigma = \sqrt{\lambda}$$

